Precipitation-evapotranspiration index sensitivity to choice of probability distribution in drought assessment

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Outline

What is Drought?

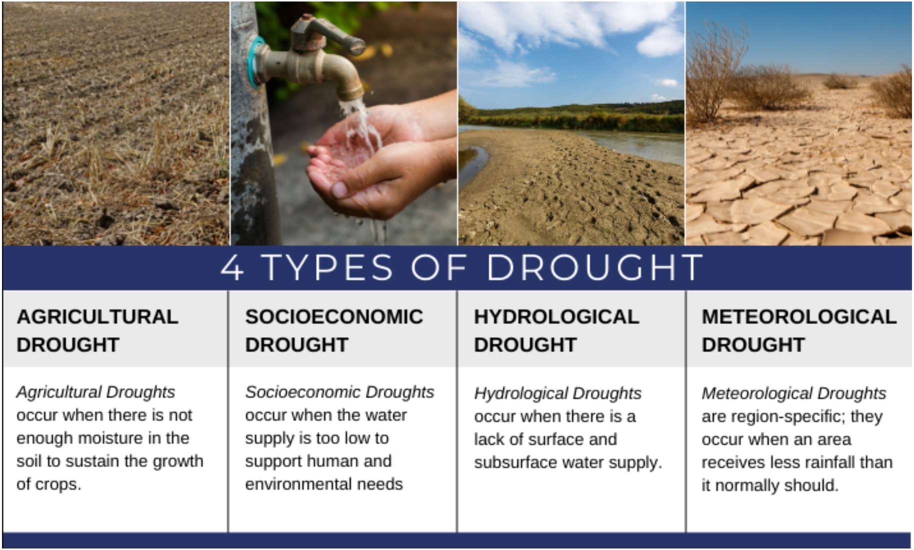
Definitions

- Drought is a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is a slow-onset disaster characterized by the lack of precipitation, resulting in a water shortage. (WHO)
- A drought is a period when an area or region experiences below-normal precipitation. The lack of adequate precipitation, either rain or snow, can cause reduced soil moisture or groundwater, diminished stream flow, crop damage, and a general water shortage. (National Geographic)

Drought

- **DURATION:** A drought can last for months or years or may be declared in the early days if low rain perception is clear for some period.
- **IMPACTS:** Droughts can have a substantial impact on the ecosystem and agriculture of the affected region and harm the local population and economy.

Types of Drought



Source: https://news.tempest.earth/what-causes-drought

Types of Drought

- **METEOROLOGICAL:** Meteorological drought arises from insufficient precipitation, affecting water infiltration, runoff, and groundwater recharge.

Source: https://www.shutterstock.com/image-photo/big-tree-on-drought-land-dry-1606270384

Causes of Drought

- Human Causes
  - Climate change
  - Deforestation
  - Agriculture
  - High water demand
- Natural Causes
  - Changes in ocean temperatures
  - The jet stream

Source: https://scijinks.gov/what-causes-a-drought/

Drought in Baluchistan

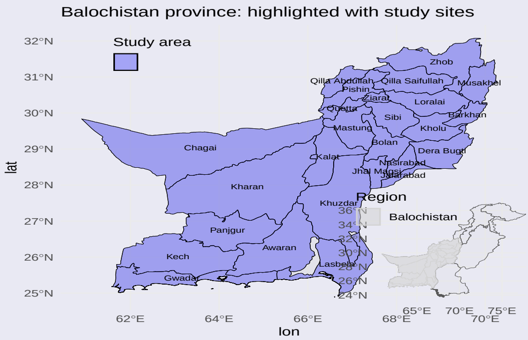
- Historical Droughts
  - Baluchistan province has faced many severe droughts in the past, i.e.,1967 to 1969, 1971, 1973 to 1975, 1994, and 1998 to 2002, which had a drastic impact on livelihoods and its economy. These prolonged droughts destroyed nearly 80% of fruit orchards and around two million animals died.
  - These assessments and measures are according to the case studies by government, different environmentalist organization and United Nations Development Program (UNDP) case studies.
  - The history shows that the drought effects on Baluchistan are mainly due to climate change happening around the world and most importantly happening in the Pakistan itself.

Motivation of the Study

- Safeguarding water and food security in Baluchistan.
- Protecting livelihoods from climate change impacts.
- Understanding and mitigating drought phenomena.
- Empowering decision-makers with knowledge and tools.
- Ensuring well-being of present and future generations.

Methodology

Study Area



Study Area

- Weather station data sourced from the NASA website https://power.larc.nasa.gov/data-access-viewer.
- Data comprises daily aggregate precipitation (in millimeters) and temperature (in celsius) for the province Balochistan.
- Data covers 40 years from January 1st, 1985, to December 31st, 2024.

Potential Evapotranspiration (PET) Hargreaves Method

- Potential Evapotranspiration (PET) represents the amount of water that would be lost to the atmosphere through evaporation and plant transpiration if sufficient water were available.
- PET is a key input in drought indices because it reflects atmospheric demand for moisture.
- The Hargreaves PET can be expressed as

$$\text{PET} = 0.0023\, R_a\, (T_{\text{mean}} + 17.8)\, (T_{\max} - T_{\min})^{0.5} \tag{1}$$

- Widely applied in drought and climate studies where Penman Monteith data are unavailable.
- Performs well in data-scarce regions because it does not require humidity, wind speed, or surface radiation measurements.

Standardized Precipitation Evapotranspiration Index

- SPEI is a drought index based on climatic data.
- SPEI series is constructed using GLO, GEV, and BAT models.

Table: Classification of drought levels, based on SPEI values

SPEI value	Drought category
SPEI ≥ 2.00	Extremely humid
$1.50 \leq$ SPEI ≤ 2.00	Very humid
$1.00 \leq$ SPEI ≤ 1.50	Humid
$-1.00 \leq$ SPEI ≤ 1.00	Normal
$-1.50 \leq$ SPEI ≤ -1.00	Dry
$-2.00 \leq$ SPEI ≤ -1.50	Very dry
SPEI ≤ -2.00	Extremely dry

Source: Labedzki, L. (2006). Susze rolnicze. Zarys problematyki oraz metody monitorowania i klasyfikacji. Woda. Srodowisko. Obszary Wiejskie. Rozprawy Naukowe i Monografie, (17):1–107.

Generalized Logistic (GLO) distribution

- GLO distribution models hydrological extremes.
- The CDF of GLO is written as

$$F(x;\mu,\sigma) = \frac{1}{1+e^{-\frac{x-\mu}{\sigma}}}, \quad -\infty < x < \infty \tag{2}$$

- The log-likelihood function for unknown parameters vector $\boldsymbol{\theta} = (\mu, \sigma)$ is defined as

$$l(\boldsymbol{\theta}) = \sum_{k=1}^{K} \left[-\log\sigma - \frac{x_k-\mu}{\sigma} - 2\log\left(1+e^{-\frac{x_k-\mu}{\sigma}}\right)\right] \tag{3}$$

Generalized Extreme Value (GEV) distribution

- GEV distribution models the block maxima.
- The inverse CDF of GEV is written as

$$G_{\alpha,\beta,\kappa}(x) = \begin{cases} \exp\left\{-\left[1+\kappa\left(\frac{x-\alpha}{\beta}\right)\right]^{-\frac{1}{\kappa}}\right\} & \text{if } \kappa \neq 0, \kappa\left(\frac{x-\alpha}{\beta}\right) > 0 \\ \exp\left\{\exp\left(\frac{x-\alpha}{\beta}\right)\right\} & \text{if } \kappa = 0, x \in \mathbb{R}. \end{cases} \quad (4)$$

- The log-likelihood function for unknown parameters vector $\boldsymbol{\theta} = (\alpha, \beta, \kappa)$ is defined as

$$\text{l}(\boldsymbol{\theta}) = -n\log\beta - \left(1+\frac{1}{\kappa}\right)\sum_{i=1}^{n}\log\left[1+\kappa\frac{(x_i-\alpha)}{\beta}\right] - \sum_{i=1}^{n}\left[1+\kappa\frac{(x_i-\alpha)}{\beta}\right]^{-\frac{1}{\kappa}} \quad (5)$$

BULK-AND-TAILS (BATs) distribution

- BATs distribution models SPEI distribution with flexible lower and upper tails.
- The BATs model is specified by its cumulative distribution function (CDF) as

$$\mathcal{F}_\theta(y) = \mathcal{T}_\nu(\mathcal{H}_\theta(y)), \tag{6}$$

- Using the standard logistic distribution:

$$\mathcal{G}(y) = e^y/1 + e^y, \tag{7}$$

we get

$$\Upsilon(y) = \log(1 + e^y) \tag{8}$$

- The BATs inner CDF transformation is defined as:

$$\mathcal{H}_\theta(y) = \left[1 + \gamma_2\, \Upsilon\left(\frac{y - \alpha_2}{\beta_2}\right)\right]^{1/\gamma_2} - \left[1 + \gamma_1\, \Upsilon\left(\frac{\alpha_1 - y}{\beta_1}\right)\right]^{1/\gamma_1} \tag{9}$$

Risk Assessment

- Drought risk is the likelihood of experiencing a drought.
- The risk is defined as

$$R_{\mathcal{D}}(G_{\alpha,\beta,\kappa}, d_{cr}) = \mathcal{D} \tag{10}$$

and

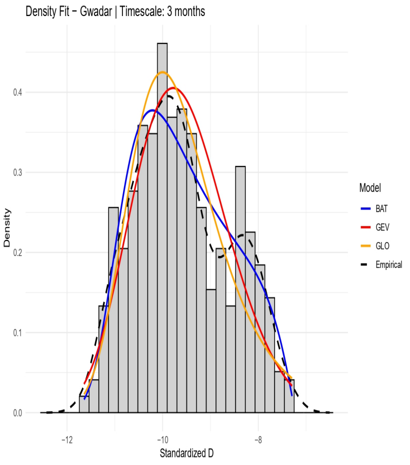
$$R_{\mathcal{D}_{\mathcal{Q}}}(G^{-1}_{\alpha,\beta,\kappa}, p_{cr}) = \mathcal{D}_Q, \tag{11}$$

where $G^{-1}_{\alpha,\beta,\kappa}$ is inverse CDF of GEV distribution.

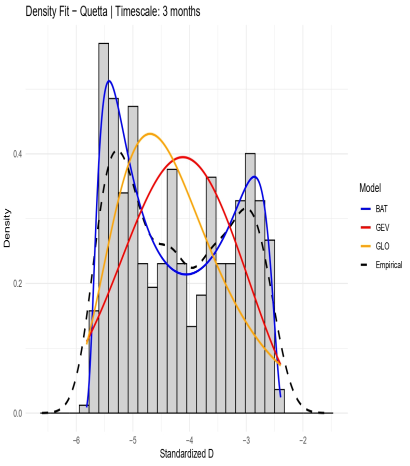
Source: Piwowar, A. and Ku zmi nski, L. (2023). Drought risk probabilistic models based on extreme value theory. Environmental Science and Pollution Research, 30(42):95945–95958.

Density Plots

Density Plots for Timescale 3 Months

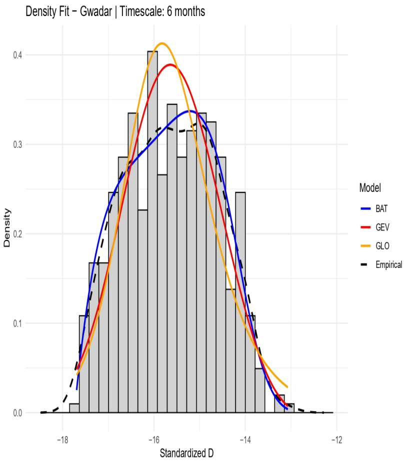


Gawadar Timescale 3 Month

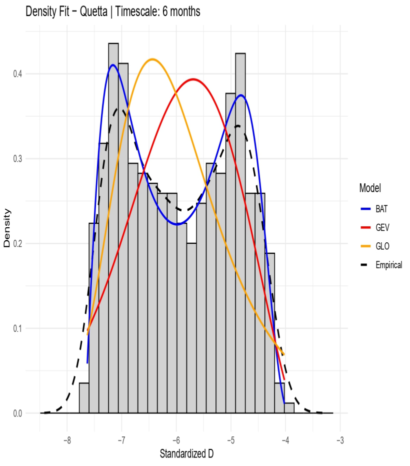


Quetta Timescale 3 Month

Density Plots for Timescale 6 Months

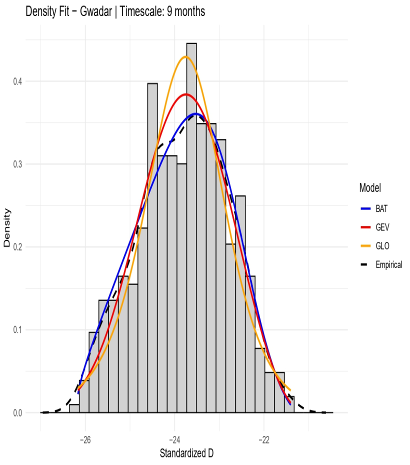


Gawadar Timescale 6 Month

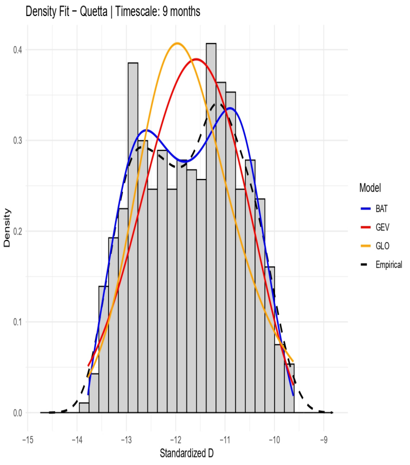


Quetta Timescale 6 Month

Density Plots for Timescale 9 Months

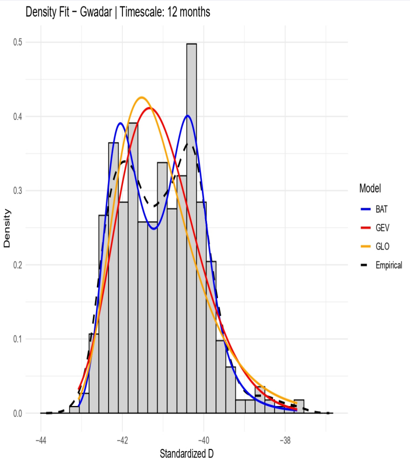


Gawadar Timescale 9 Month

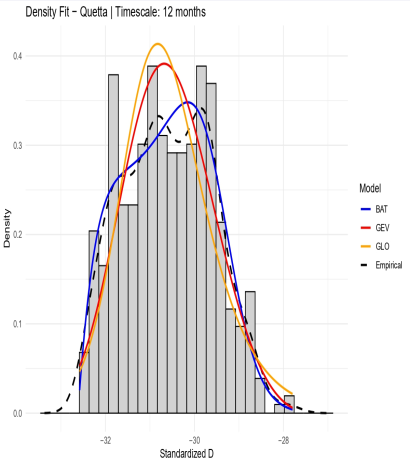


Quetta Timescale 9 Month

Density Plots for Timescale 12 Months

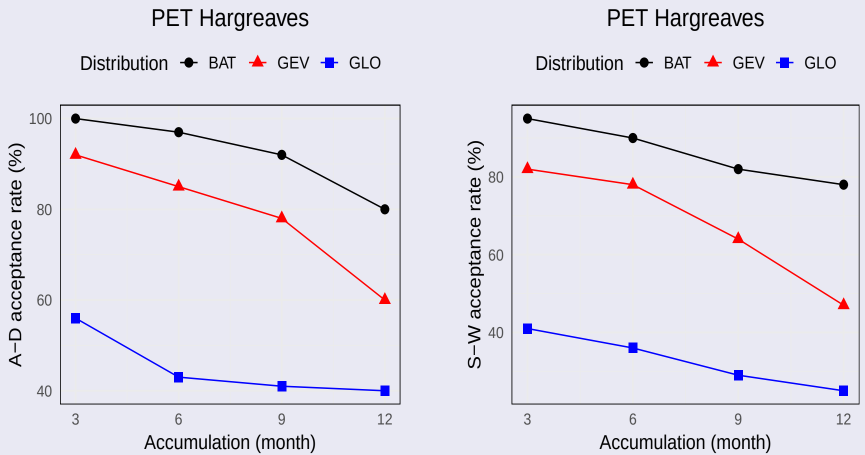


Gawadar Timescale 12 Month



Quetta Timescale 12 Month

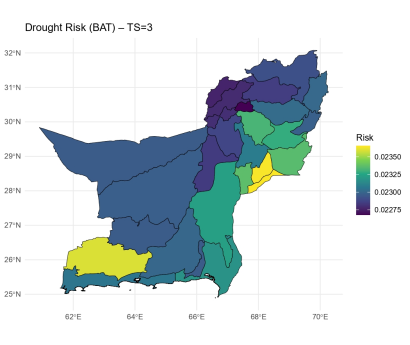
Goodness of Fit Test



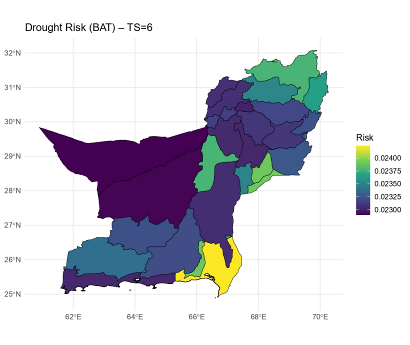
Acceptance rate for a) Anderson Darling and b) Shapiro Wilk Tests

Drought Risk

Drought Risk for Timescales 3 and 6 Months

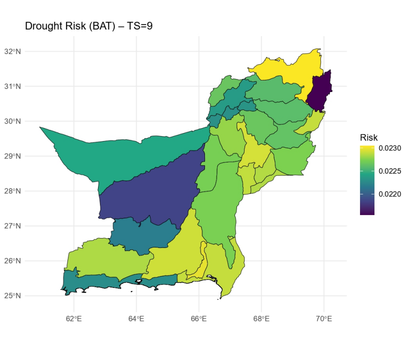


BAT Timescale 3 Month

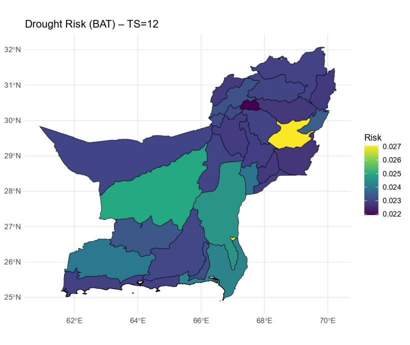


BAT Timescale 6 Month

Drought Risk for Timescales 9 and 12 Months



BAT Timescale 9 Month



BAT Timescale 12 Month

Conclusion

Conclusion

- The proposed framework effectively measures drought conditions even with extreme climatic observations.
- BATs distribution provides the most reliable modeling of drought behavior across timescales.
- Findings suggest that long-term climate change has altered the region's drought risk profile.
- The choice of probability distribution significantly influences SPEI-based drought assessment.

Future Work

Work in progress

- Extend analysis by evaluating additional probability distributions.
- Incorporate climatic and regional covariates to improve model accuracy.
- Enhance the spatial representation of drought risk with more detailed inputs.
- Continue refining the framework as the study progresses.

THANK YOU

Any Questions?